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MPNNLight: A Self-Attention Enhanced Message Passing Graph Neural Network for Multi-Intersection Traffic Signal Control

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Abstract

Coordinated control of multiple intersections remains a major challenge in urban traffic systems due to complex local–global dependencies. To address this issue, the MPNNLight framework is proposed, a message-passing graph neural network for cooperative multi-intersection traffic signal control. The core innovation of this work is the proposed Attention-Wavelet Spatial Transformer (AWSformer), a message computation module that combines self-attention mechanisms and 2D discrete wavelet transform (2D-DWT) to adaptively compute inter-agent messages across spatial and frequency domains. This design enables efficient extraction of multi-scale dependencies and enhances coordination among intersections. Experiments on diverse datasets demonstrate that MPNNLight achieves superior performance and efficiency, compared with existing GNN-based traffic signal control methods.

Keywords: graph neural network; traffic signal control; message passing; wavelet transform

1. Introduction

With the rapid pace of urbanization and the continuous increase in vehicle ownership, traffic congestion has become a pervasive challenge in modern cities. Intersections, as the key nodes where different traffic streams converge, play a central role in determining network-wide traffic performance. While traditional single-intersection signal control strategies can locally improve flow efficiency, they often fail to capture the spatial correlations and dynamic interactions among adjacent intersections. Consequently, multi-intersection traffic signal control (TSC) has emerged as a vital research direction, aiming to achieve coordinated and adaptive control across an entire network. By enabling cooperative signal optimization, such approaches can significantly enhance overall traffic efficiency, reduce vehicle delays, and promote greener, more sustainable urban transportation systems.

Early studies on traffic signal control mainly relied on rule-based or optimization-based methods, such as fixed-time control, actuated control, and model-based optimization techniques. Although these approaches can provide stable signal plans under relatively predictable traffic conditions, they typically require accurate traffic models and extensive manual parameter tuning. Moreover, their ability to adapt to rapidly changing traffic patterns is limited, especially in complex urban environments with stochastic demand and dynamic traffic flows.



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To address these limitations, reinforcement learning (RL) has recently gained significant attention in the traffic signal control community. RL-based TSC methods enable agents to learn adaptive signal control policies directly from traffic observations through interaction with the environment, without requiring explicit traffic flow models. By continuously optimizing control policies based on reward feedback, RL algorithms can effectively adapt to dynamic traffic conditions and improve long-term traffic efficiency. A variety of RL algorithms have been applied to traffic signal control, including Q-learning, deep Q-networks, and actor-critic methods. In particular, advanced off-policy algorithms such as Soft Actor-Critic (SAC) [1] provide stable training and improved exploration capabilities, making them well suited for large-scale traffic control problems.

In recent years, numerous studies have been conducted on multi-intersection traffic signal control, exploring various optimization strategies and learning-based approaches to improve network-wide coordination and efficiency. Graph neural networks (GNNs) [2–5] have been widely applied in multi-intersection traffic signal control [6–9]. By representing intersections as nodes and road connections as edges, GNN-based models naturally capture the topological structure of urban road networks and enable information exchange among neighboring intersections. This graph-based formulation allows traffic controllers to learn spatial dependencies and coordination strategies across multiple intersections, significantly improving the scalability and effectiveness of multi-agent traffic control systems.

Despite these promising developments, existing GNN-based TSC approaches still face several limitations. First, many methods rely on Graph Convolutional Networks (GCNs), which aggregate information from neighboring nodes using fixed convolutional operators. While effective for capturing local spatial relationships, such aggregation schemes are often limited in modeling long-range dependencies and complex interactions across distant intersections. Second, Graph Attention Networks (GATs) introduce attention mechanisms to adaptively weigh information from neighboring nodes, improving flexibility in message aggregation. However, GAT-based models often suffer from over-smoothing and computational overhead when applied to large-scale traffic networks. Third, most existing approaches primarily focus on local neighborhood interactions and fail to sufficiently incorporate global graph-level information. As a result, these models may struggle to coordinate traffic control decisions across multiple intersections when network-wide traffic conditions change significantly.

To overcome these challenges, more expressive graph learning frameworks are required to effectively capture both local and global traffic dynamics. Message Passing Neural Networks (MPNNs) [10] provide a flexible and general framework for graph representation learning, where node representations are iteratively updated through message exchange between connected nodes. Compared with conventional GCN and GAT architectures, MPNN allows more flexible message computation and aggregation mechanisms, enabling the modeling of richer spatial interactions and long-range dependencies across the traffic network.

Motivated by these observations, this paper proposes MPNNLight, a multi-intersection traffic signal control framework trained with the Soft Actor-Critic (SAC) reinforcement learning algorithm [1]. Compared with GAT and GCN, the Message Passing Neural Network (MPNN) framework is more suitable for multi-intersection control because it enables flexible message propagation and effectively captures long-range spatial dependencies across intersections. MPNNLight integrates both global and local traffic representations to achieve cooperative decision-making among intersections. The global information is obtained through a newly proposed AWSformer module, which serves as the message computation function within the MPNN architecture. As the key innovation of MPNNLight, AWSformer combines the self-attention mechanism with the 2D discrete wavelet transform (2D-DWT) [11] to adaptively compute inter-agent messages across spatial and frequency

domains, enhancing multi-scale feature representation and coordination ability. Experimental results on eight benchmark datasets demonstrate that MPNNLight outperforms existing graph neural network-based traffic signal control models, such as those built on GAT, GCN, and Graph Transformer [12] architectures.

The main contributions of this work are summarized as follows:

- MPNNLight, a novel graph-based reinforcement learning framework for multi-intersection traffic signal control, is proposed. It leverages message passing to effectively capture spatial dependencies and cooperative interactions across intersections.
- A new message computation module, AWSformer, is introduced, which integrates self-attention mechanisms with 2D discrete wavelet transform (2D-DWT) to extract multi-scale spatial–frequency traffic features and improve the quality of information exchange among intersections.
- A cooperative learning framework is designed that combines the expressive power of MPNN with the stability of the Soft Actor-Critic (SAC) reinforcement learning algorithm, enabling efficient training and adaptive traffic signal control in dynamic urban environments.
- Extensive experiments on eight synthetic benchmark datasets and five real-world datasets demonstrate that the proposed method consistently outperforms several representative traffic signal control approaches based on GCN, GAT, and Graph Transformer, achieving significant improvements in both average travel time and vehicle waiting time.

The remainder of this paper is organized as follows. Section 2 reviews related work on reinforcement learning for traffic signal control. Section 3 presents the proposed MPNN-Light framework, including the AWSformer-based message passing mechanism and the overall model architecture. Section 4 describes the experimental setup and evaluates the performance of the proposed method on both synthetic and real-world datasets. Finally, Section 5 concludes the paper and discusses potential directions for future work.

2. Related Work

2.1. Foundational Graph-Based Reinforcement Learning for TSC

The shift toward multi-intersection coordination has been driven by the integration of graph neural networks (GNNs) with Multi-Agent Reinforcement Learning (MARL). Early foundational work by Wei et al. established a graph-based framework enabling cooperative control through inter-agent communication and shared state representations [13]. Building on this, Devailly et al. introduced the Inductive Graph Reinforcement Learning (IG-RL) framework, utilizing Graph Convolutional Networks (GCNs) to learn policies that generalize to new road networks without retraining, significantly enhancing scalability [14]. To further address the complexities of large-scale urban grids, Liu et al. proposed GlobalLight, which balances local dependencies via multi-head Graph Attention Networks (GATs) while exploiting global influence through parameter sharing among distant but similar intersections [15]. These pioneering studies demonstrated that graph-structured representations are superior to independent RL agents in capturing the inherent connectivity of urban traffic.

2.2. Advanced Spatial–Temporal and High-Order Modeling

Recent research has pushed the boundaries of spatial–temporal modeling to capture more nuanced traffic dynamics. Wang et al. introduced a framework incorporating spatio-temporal hypergraphs into the critic network to model high-order correlations among multiple intersections [16]. Similarly, contemporary directed hypergraph learning frameworks (e.g., Directed Hypergraph-based Proximal Policy Optimization, DHG-PPO) have been developed to extract complex spatio-temporal attributes beyond simple pairwise connections [17].

Recent efforts have also explored physics-guided spatio-temporal modeling. Pan et al. [18] incorporate traffic flow priors into STGNNs to improve interpretability and robustness, yet such methods still operate mainly in the time domain without explicitly modeling multi-scale frequency characteristics.

Innovations in attention mechanisms have further refined these models. For instance, combining short-term traffic flow prediction with GATs enables proactive regional signal decisions [19], while advanced spatio-temporal attention mechanisms allow agents to dynamically weight neighboring states in large-scale networks [20]. To enhance the granularity of these representations, researchers have introduced channel attention for node modeling [21] and utilized restart random walks within GATs to provide a comprehensive global perspective of the network topology [22]. Furthermore, recent adaptive signal control methods based on GNNs and dynamic entropy-constrained Soft Actor-Critic (SAC) have shown promise in extracting both regional and local traffic features [23]. However, while these models excel at capturing spatial dependencies, they predominantly operate in the spatial domain, often struggling to isolate multi-scale traffic variations, such as sudden high-frequency local congestion versus smooth low-frequency trends.

2.3. Generalization, Scalability, and Cross-City Transferability

Achieving zero-shot generalization remains a primary objective in the field. Beyond the inductive capabilities of IG-RL [14], recent frameworks like TransferLight utilize hierarchical heterogeneous directed GNNs to achieve deployment across varying intersection topologies [24]. Transformer architectures have also emerged as a robust alternative; X-Light employs a stacked Transformer architecture to aggregate state–action–reward trajectories for cross-city transferability [25]. Similarly, Centralized Collaborative Graphical Transformers (CCGNs) have been designed to facilitate signal-free corridors by capturing long-range dependencies more effectively than standard GCNs [26]. To mitigate performance degradation under shifting traffic distributions, model-based reinforcement learning with policy reuse has also been proposed [27]. Despite these strides, the heavy reliance on global self-attention or deep Transformer architectures often incurs prohibitive computational overhead, limiting real-time deployability.

2.4. Real-World Constraints and Computational Efficiency

Current research also addresses physical constraints and training efficiency. Constrained MARL approaches, such as Multi-Agent Proximal Policy Optimization with Lagrange Cost Estimator (MAPPO-LCE), treat signal control as a constrained optimization problem to balance throughput and safety [28]. Other studies explore heterogeneous GNNs for the joint coordination of vehicle platooning and signal control [29]. On the efficiency front, methods like multi-scene aggregation convolutional learning (MacLight) [30] and temporal difference-aware graph convolutional frameworks (TeDA-GCRL) [31] bypass the computational bottlenecks of standard message-passing, offering faster convergence. Comprehensive reviews [32,33] and studies on traffic pattern recognition [34] emphasize that while GNNs have revolutionized TSC, they fail to systematically decouple frequency-domain components. This gap motivates the proposed MPNNLight architecture, which utilizes AWSformer and 2D discrete wavelet transforms to achieve multi-scale feature representation across both spatial and frequency domains.

3. Method

In this section, we describe the overall architecture and operation of the proposed MPNNLight algorithm, as illustrated in Figure 1. Each intersection is managed by an independent agent whose input includes local traffic features of the intersection and

global structural features captured by a graph neural network. Integrating these enables coordinated control across intersections.

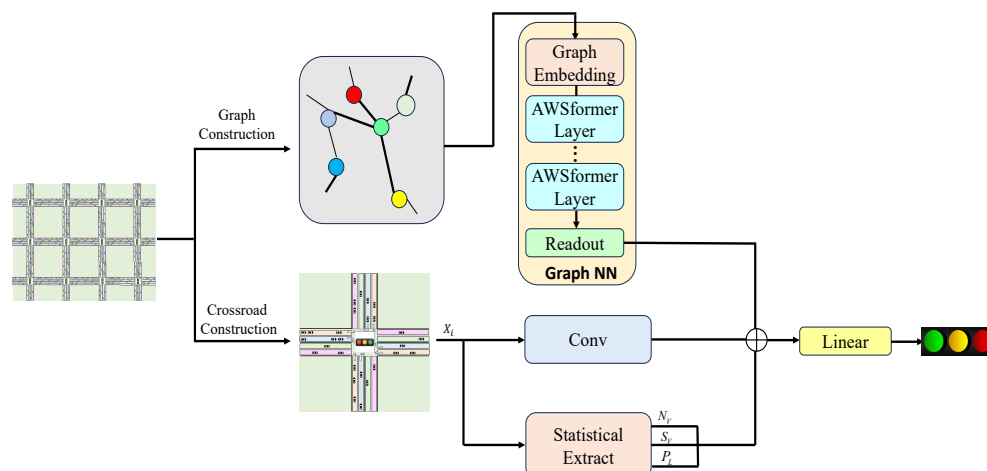


Figure 1. Overall architecture of MPNNLight.

For the i -th intersection, the vehicle state matrix X_i contains position and velocity information of vehicles on connected lanes. A convolutional network $f_{\text{conv}}(\cdot)$ extracts spatial features:

$$c_i = f_{\text{conv}}(X_i), \quad (1)$$

Meanwhile, aggregated statistics according to X_i form an auxiliary state vector:

$$s_i = [N_i^V, S_i^V, P_i^L], \quad (2)$$

where N_i^V is the vehicle count on the incoming lanes, S_i^V denotes the average lane speed and P_i^L denotes the current signal phase of the intersection. The local representation is obtained by concatenating c_i and embedded statistical features:

$$h_i = [c_i \parallel \text{LN}(s_i)], \quad (3)$$

At the global level, the entire traffic network is modeled as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the node set $\mathcal{V}$ corresponds to all intersections, and the edge set $\mathcal{E}$ represents the connecting roads between them. The node features of each intersection are constructed from its local traffic state:

$$f_i^{(0)} = [N_i^V, S_i^V], \quad (4)$$

where $f_i^{(0)}$ denotes the initial node feature vector of intersection i , encapsulating its fundamental traffic conditions. As defined similarly in Equation (2), N_i^V represents the number of vehicles on the incoming lanes, and S_i^V denotes the average lane speed. The edge features are further defined to characterize the traffic attributes of the connecting roads:

$$e_{ij} = [L_{ij}, S_{ij}, N_{ij}], \quad (5)$$

where L_{ij} represents the road length between intersections i and j , S_{ij} denotes the average road speed, and N_{ij} indicates the number of vehicles traveling on that road segment.

Specifically, in the constructed graph structure, a message passing neural network (MPNN) is employed to facilitate state interaction among multiple intersections. During

the message passing process at layer t , each node i receives messages from its neighboring nodes $j \in \mathcal{N}(i)$:

$$m_{ij}^{(t)} = \psi(f_i^{(t)}, f_j^{(t)}, e_{ij}), \quad (6)$$

where $\psi(\cdot)$ denotes a learnable message function that models the interaction between node states and edge attributes. To enhance the expressive capacity of $\psi(\cdot)$, the MPNNLight algorithm introduces a novel attention- and wavelet-based learnable message function module, termed **AWSformer**, as illustrated in Figure 2. The core idea of AWSformer is to integrate edge feature fusion and 2D-DWT into the attention mechanism, enabling dual-stage spatial–frequency attention that captures global patterns while remaining sensitive to local variations.

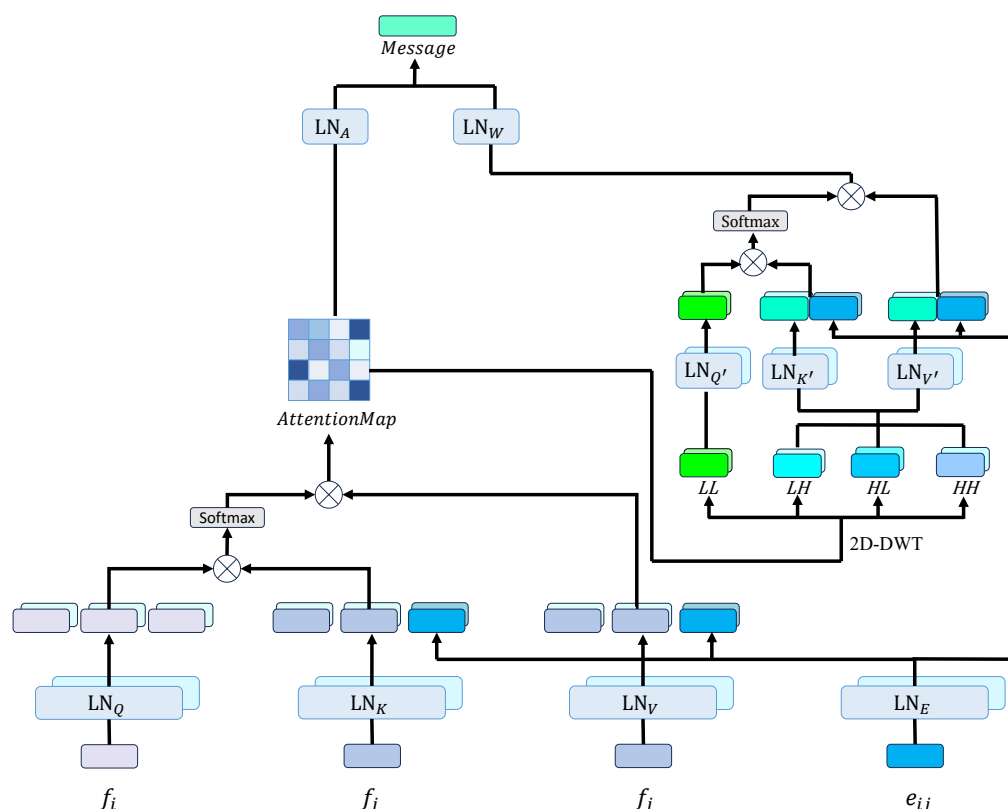


Figure 2. The structure of AWSformer.

In the t -th layer of AWSformer, for any pair of adjacent nodes i and j connected by edge feature e_{ij} , the node information $f_i^{(t)}$ and $f_j^{(t)}$ represent the local traffic states of intersections, whereas the edge embedding e_{ij} characterizes the topological and flow attributes of the connecting road (including road length, average speed, and vehicle count). First, the node and edge features are linearly embedded as follows:

$$\begin{aligned} Q_i &= \text{LN}_Q(f_i^{(t)}), & K_j &= \text{LN}_K(f_j^{(t)}), \\ V_j &= \text{LN}_V(f_j^{(t)}), & E_{ij} &= \text{LN}_E(e_{ij}), \end{aligned} \quad (7)$$

where Q_i denotes the query representation of intersection i , characterizing the current traffic demand of the target node. K_j and V_j denote the key and value representations of neighboring intersection j , where K_j is used to evaluate how relevant the traffic condition of node j is to node i , and V_j carries the traffic information of node j for message aggregation. E_{ij} denotes the embedding of the edge feature associated with the road segment con-

necting node j to node i . $\text{LN}_Q(\cdot)$, $\text{LN}_K(\cdot)$, $\text{LN}_V(\cdot)$, and $\text{LN}_E(\cdot)$ denote the corresponding linear layers.

To explicitly encode the semantic information of roads during the attention computation, the edge features are concatenated into the key and value representations:

$$\tilde{K}_j = \text{Concat}(K_j, E_{ij}), \quad \tilde{V}_j = \text{Concat}(V_j, E_{ij}), \quad (8)$$

This joint node–edge modeling strategy enables the model to perform information exchange not only based on intersection-level states but also by incorporating the characteristics of connecting roads, thereby capturing topological dependencies and traffic flow propagation patterns across the network. More specifically, integrating node representations with road attributes allows the attention mechanism to adaptively weight information from neighboring intersections according to the properties of the connecting roads. In real traffic networks, interactions between intersections are not solely determined by their local states, but are also strongly influenced by the structural and operational characteristics of the roads that connect them, such as road length, capacity, speed limits, and directional flow constraints. These factors determine how efficiently vehicles propagate between intersections and therefore directly affect the strength and relevance of inter-node interactions. By concatenating node features with corresponding edge features when computing the cross-attention, the model explicitly conditions the attention weights on both intersection states and road attributes. As a result, the attention mechanism can learn to emphasize messages transmitted through critical or high-capacity links while suppressing less informative or weakly connected interactions. This design also introduces a form of topology-aware relational bias, enabling the model to better reflect the physical traffic network structure and the directional propagation of congestion and vehicle flows. Consequently, the joint node–edge attention formulation improves the ability of the message passing process to capture realistic traffic dependencies and enhances the effectiveness of network-level coordination. Subsequently, for each attention head $\tilde{h} \in \{1, \dots, H\}$, the spatial attention weight between node i and its neighbor $j \in \mathcal{N}(i)$ is computed as

$$A_{ij}^{(\tilde{h})} = \frac{\exp\left(\frac{Q_i^{(\tilde{h})}(\tilde{K}_j^{(\tilde{h})})^\top}{\sqrt{d}}\right)}{\sum_{j' \in \mathcal{N}(i)} \exp\left(\frac{Q_i^{(\tilde{h})}(\tilde{K}_{j'}^{(\tilde{h})})^\top}{\sqrt{d}}\right)}, \quad (9)$$

where d denotes the feature dimension of the query and key in each attention head. Based on these head-wise attention coefficients, the corresponding attended message is obtained as

$$m_{ij}^{(\tilde{h})} = A_{ij}^{(\tilde{h})} \tilde{V}_j^{(\tilde{h})}, \quad m_{ij}^{(\tilde{h})} \in \mathbb{R}^C, \quad (10)$$

where C denotes the output feature dimension of each head after value projection. The spatial-domain message representation is then constructed by stacking the attended features from all heads:

$$M_{sp} = \text{Stack}(m_{ij}^{(1)}, m_{ij}^{(2)}, \dots, m_{ij}^{(H)}) \in \mathbb{R}^{H \times C}, \quad (11)$$

where H is the number of attention heads. In this way, M_{sp} forms a two-dimensional attention feature map, in which the first dimension corresponds to different attention heads and the second dimension corresponds to the feature channels within each head.

To achieve multi-scale decomposition of traffic features, a 2D-DWT is applied to M_{sp} to extract frequency-domain features. The fundamental idea of 2D-DWT is to perform low-pass and high-pass filtering along both the row and column directions, followed

by down-sampling. Let $M_{sp} \in \mathbb{R}^{H \times C}$ denote the spatial-domain feature matrix, and let $h_L[n]$ and $h_H[n]$ represent the low-pass and high-pass filters corresponding to the selected wavelet basis, respectively. The four sub-bands of the 2D-DWT are then defined as:

$$\begin{aligned} M_{sp,LL}[p,q] &= \sum_m \sum_n M_{sp}[m,n] h_L[2p-m] h_L[2q-n], \\ M_{sp,LH}[p,q] &= \sum_m \sum_n M_{sp}[m,n] h_L[2p-m] h_H[2q-n], \\ M_{sp,HL}[p,q] &= \sum_m \sum_n M_{sp}[m,n] h_H[2p-m] h_L[2q-n], \\ M_{sp,HH}[p,q] &= \sum_m \sum_n M_{sp}[m,n] h_H[2p-m] h_H[2q-n], \end{aligned} \quad (12)$$

where (p, q) denotes the spatial coordinates after down-sampling. Here, $M_{sp,LL}$ represents the low-frequency component obtained through low-pass filtering in both directions, which concentrates the slowly varying and globally consistent patterns in M_{sp} , such as network-level congestion trends and the overall smooth evolution of traffic states. In contrast, $M_{sp,LH}$ is produced by low-pass filtering along rows and high-pass filtering along columns, thereby emphasizing column-direction discontinuities and vertically oriented local variations (e.g., localized changes caused by lane-imbalance or upstream-to-downstream propagation along one direction); $M_{sp,HL}$ is generated by high-pass filtering along rows and low-pass filtering along columns, capturing row-direction fluctuations and horizontally oriented perturbations; and $M_{sp,HH}$ applies high-pass filtering in both directions, retaining fine-grained details, sharp transitions, and abrupt local events such as sudden queue spillback, bursty arrivals, or phase-switch-induced transients. We employ the 2D-DWT at this stage for three main reasons. First, traffic interactions encoded in the spatial-domain message M_{sp} naturally contain multi-scale structures: global coordination cues evolve smoothly over space/time, while local disturbances can be sparse and intermittent. By decomposing M_{sp} into $M_{sp,LL}$ and the high-frequency sub-bands $\{M_{sp,LH}, M_{sp,HL}, M_{sp,HH}\}$, the model can explicitly separate coarse global dynamics from directional local variations, which facilitates more informative and disentangled message representations. Second, the sub-bands provide complementary inductive bias: $M_{sp,LL}$ supplies stable global context for coordination, whereas $\{M_{sp,LH}, M_{sp,HL}, M_{sp,HH}\}$ preserve anisotropic (direction-dependent) changes that are closely related to road topology and directional flows, enabling the subsequent module to attend to the most relevant local perturbations without being dominated by global trends. Third, 2D-DWT improves robustness to high-frequency noise in M_{sp} . Since noisy fluctuations and measurement jitters often appear in high-frequency regions, isolating them into $\{M_{sp,LH}, M_{sp,HL}, M_{sp,HH}\}$ prevents such noise from directly contaminating the globally consistent component $M_{sp,LL}$, while still keeping potentially useful sharp-event information in an explicit form. Overall, the 2D-DWT decomposes the spatial-domain message M_{sp} into complementary frequency components, enabling joint modeling of global and local dynamics, enhancing directional sensitivity, and improving robustness to high-frequency noise:

$$M_{sp} \xrightarrow{\text{2D-DWT}} \{M_{sp,LL}, M_{sp,LH}, M_{sp,HL}, M_{sp,HH}\}, \quad (13)$$

Beyond serving as a feature transformation, the design of AWSformer can also be interpreted from the perspective of traffic-flow propagation and multi-scale traffic dynamics. In urban traffic networks, the state evolution at adjacent intersections is not only spatially correlated, but also exhibits wave-like propagation phenomena, such as queue spillback, stop-and-go waves, and shockwave transmission. From a macroscopic perspective, traffic dynamics satisfy the conservation law [35]

$$\frac{\partial \rho(x, \tau)}{\partial \tau} + \frac{\partial q(x, \tau)}{\partial x} = 0, \quad (14)$$

where $\rho(x, \tau)$ and $q(x, \tau)$ denote traffic density and flow at location x and decision step τ , respectively. Since $q = Q(\rho)$ follows the fundamental diagram, linearization around an operating point ρ^* yields

$$q(x, \tau) \approx Q(\rho^*) + Q'(\rho^*)(\rho(x, \tau) - \rho^*), \quad (15)$$

which leads to the first-order wave propagation form

$$\frac{\partial \rho(x, \tau)}{\partial \tau} + c(\rho^*) \frac{\partial \rho(x, \tau)}{\partial x} = 0, \quad c(\rho^*) = Q'(\rho^*). \quad (16)$$

This indicates that traffic evolution naturally contains both smooth large-scale trends and rapidly changing local perturbations.

Under this view, the MPNN framework provides a graph-based approximation of traffic propagation over the road network, where each intersection aggregates influences transmitted from its neighboring intersections through connecting roads. In particular, the message passing mechanism models how local traffic states and road relations jointly affect downstream and upstream interactions over the graph topology. On this basis, the self-attention mechanism further enables AWSformer to adaptively determine the relative importance of different neighboring intersections and propagation paths, which is physically meaningful because the impacts of different incoming directions and adjacent roads are inherently heterogeneous under dynamic traffic conditions.

The 2D-DWT decomposition can then be interpreted as a multi-scale separation of inter-intersection traffic interactions. Specifically, the low-frequency component $M_{sp,LL}$ mainly captures slowly varying network-level coordination patterns, whereas the high-frequency components $\{M_{sp,LH}, M_{sp,HL}, M_{sp,HH}\}$ characterize directional local disturbances and abrupt transitions, such as queue buildup, directional imbalance, phase-switch-induced transients, and shockwave propagation. If the local traffic feature is decomposed as

$$f_i(\tau) = \bar{f}_i(\tau) + \delta f_i(\tau), \quad (17)$$

where $\bar{f}_i(\tau)$ denotes the smooth background component and $\delta f_i(\tau)$ denotes the perturbation component, then the low-frequency branch mainly preserves information dominated by $\bar{f}_i(\tau)$, while the high-frequency branches are more sensitive to variations dominated by $\delta f_i(\tau)$. Therefore, the design of AWSformer is not merely a heuristic combination of MPNN, self-attention, and 2D-DWT, but a traffic-oriented message computation mechanism that aligns graph-based propagation, adaptive interaction weighting, and multi-scale frequency decomposition with the physical characteristics of urban traffic evolution.

The high-frequency sub-bands are then concatenated to form new key and value representations, while the edge features are further integrated to preserve the semantic information of roads:

$$\begin{aligned} Q'_{fq} &= \text{LN}_{Q'}(M_{sp,LL}), \\ K'_{fq} &= \text{LN}_{K'}([M_{sp,LH}, M_{sp,HL}, M_{sp,HH}]), \\ V'_{fq} &= \text{LN}_{V'}([M_{sp,LH}, M_{sp,HL}, M_{sp,HH}]), \end{aligned} \quad (18)$$

$$\tilde{K}'_{fq} = \text{Concat}(K'_{fq}, E_{ij}), \quad \tilde{V}'_{fq} = \text{Concat}(V'_{fq}, E_{ij}) \quad (19)$$

The frequency-domain attention weights and corresponding weighted messages are then computed as:

$$A'_{fq} = \text{Softmax}\left(\frac{Q'_{fq}(\tilde{K}'_{fq})^\top}{\sqrt{d'}}\right), \quad M_{fq} = A'_{fq} \tilde{V}'_{fq}, \quad (20)$$

Finally, the spatial-domain message M_{sp} and the frequency-enhanced message M_{fq} are each processed by independent linear layers and then concatenated to generate the final message representation:

$$m_{ij}^{(t)} = \text{Concat}(\text{LN}_A(M_{sp}), \text{LN}_W(M_{fq})), \quad (21)$$

After message generation through AWSformer, each node aggregates messages from its neighboring nodes in order to incorporate traffic information propagated from adjacent intersections. In an urban traffic network, the state of an intersection is not independent; instead, it is strongly influenced by upstream and downstream intersections through connecting roads. For example, congestion or queue buildup at one intersection may propagate along the road segment and affect the traffic conditions of its neighboring intersections. Therefore, modeling such spatial interactions is essential for coordinated traffic signal control.

Formally, for each node i , the incoming messages from all neighbors $j \in \mathcal{N}(i)$ are aggregated as

$$m_i^{(t)} = \bigoplus_{j \in \mathcal{N}(i)} m_{ij}^{(t)}, \quad (22)$$

where $\mathcal{N}(i)$ denotes the set of neighboring intersections connected to node i , and $\bigoplus$ represents the aggregation operator. In this work, summation is adopted as the aggregation function due to its permutation-invariant property and its ability to accumulate traffic influence from multiple neighboring intersections. This operation allows node i to collect traffic flow information propagated through connected roads, thereby capturing the interaction between local traffic states and surrounding network conditions.

The node representation is then updated as

$$f_i^{(t+1)} = \phi(f_i^{(t)}, m_i^{(t)}), \quad (23)$$

where $\phi(\cdot)$ is a linear update function that combines the current node state $f_i^{(t)}$ with the aggregated message $m_i^{(t)}$. From a traffic control perspective, this update step enables each intersection to refine its state representation by incorporating both its own local observations and the traffic conditions of neighboring intersections. Such integration helps the model better capture traffic propagation dynamics, including queue spillback effects and upstream–downstream vehicle interactions.

By stacking multiple message passing layers, the receptive field of each node gradually expands. After K layers, the node representation $f_i^{(K)}$ implicitly encodes traffic information from multi-hop intersections in the road network. This multi-hop propagation mechanism is particularly important for coordinated signal control, since congestion patterns and traffic waves often spread across several intersections rather than being confined to a single junction.

Finally, to capture the overall traffic condition of the entire road network, a global representation is obtained through max-pooling over all node embeddings

$$g_{\text{global}} = \text{MAXPOOL}\left(\{f_i^{(K)} \mid v_i \in \mathcal{V}\}\right), \quad (24)$$

The global embedding g_{global} summarizes the most salient traffic features across all intersections and provides a compact representation of network-level congestion patterns. In the context of multi-intersection traffic signal control, this global context complements local intersection states and enables each agent to make decisions that consider both local traffic conditions and the global coordination requirements of the entire network.

In the decision stage, the global embedding is concatenated with local features in order to integrate both network-level context and intersection-level traffic conditions:

$$u_i = [h_i || g_{\text{global}}], \quad (25)$$

where h_i denotes the local traffic representation of intersection i , integrating the convolutional feature extracted from the vehicle-state matrix and the embedded statistical traffic state, including the vehicle count, average speed, and current signal phase, while g_{global} summarizes the overall traffic condition of the entire road network. By concatenating these two representations, the decision module allows each intersection to make control decisions based not only on its local observations but also on the broader network context. This design is particularly important in multi-intersection traffic signal control, where the optimal signal strategy often depends on both local congestion levels and upstream–downstream traffic propagation patterns across the network.

The fused feature u_i is then fed into a decision layer to produce the phase-selection logits for intersection i :

$$z_i = \text{LN}_U(u_i), \quad (26)$$

where $z_i \in \mathbb{R}^4$ denotes the unnormalized scores of the four candidate signal phases. A softmax operation is further applied to obtain the corresponding policy distribution:

$$\pi(a_i | u_i) = \frac{\exp(z_{i,a_i})}{\sum_{a'_i \in \mathcal{A}_{\text{loc}}} \exp(z_{i,a'_i})}, \quad (27)$$

where $\mathcal{A}_{\text{loc}} = \{1, 2, 3, 4\}$ denotes the local action space consisting of four traffic signal phases for each intersection.

In this work, the four actions correspond to the following phase configurations: north–south through movement, north–south left-turn movement, east–west through movement, and east–west left-turn movement. Therefore, at each decision step τ , the policy network outputs a categorical distribution over the four candidate phases for intersection i .

During training, the action is sampled from the policy distribution:

$$a_{i,\tau} \sim \pi(\cdot | u_{i,\tau}), \quad a_{i,\tau} \in \mathcal{A}_{\text{loc}}, \quad (28)$$

which encourages exploration and enables the reinforcement learning agent to adequately interact with the traffic environment. During testing, a deterministic policy is adopted by selecting the phase with the highest probability:

$$\hat{a}_{i,\tau} = \arg \max_{a \in \mathcal{A}_{\text{loc}}} \pi(a | u_{i,\tau}). \quad (29)$$

This phase-selection strategy allows each intersection controller to directly choose the most appropriate signal phase according to the current traffic state, while remaining fully compatible with the discrete-action reinforcement learning framework.

All intersections are controlled by multiple agents, where each intersection corresponds to an individual agent. At each decision step τ , agent $i \in V$ selects its control action $a_{i,\tau}$ according to the policy distribution $\pi(a_{i,\tau} | u_{i,\tau})$, where $u_{i,\tau}$ denotes the de-

cision feature of intersection i at step τ . The actions of all intersections jointly form the network-level action

$$a_\tau = \{a_{i,\tau}\}_{i \in V}. \quad (30)$$

Accordingly, the joint policy is factorized as

$$\pi(a_\tau | o_\tau) = \prod_{i \in V} \pi(a_{i,\tau} | u_{i,\tau}), \quad (31)$$

where o_τ denotes the joint observation of the traffic network at step τ .

To optimize the control policy, we adopt the discrete variant of Soft Actor-Critic (SAC). SAC is an off-policy reinforcement learning algorithm based on the maximum entropy framework, which aims to maximize both the expected cumulative reward and the policy entropy. The objective function is formulated as

$$J(\pi) = \sum_{\tau=0}^T \mathbb{E}_{(o_\tau, a_\tau) \sim \rho_\pi} [\gamma^\tau (r_\tau + \alpha_e \mathcal{H}(\pi(\cdot | o_\tau)))], \quad (32)$$

where $\gamma \in [0, 1)$ is the discount factor, α_e is the temperature parameter controlling the exploration–exploitation trade-off, and $\mathcal{H}(\cdot)$ denotes the entropy. To encourage cooperative control, all agents share the same global environment reward during training, while the policy and critic networks are parameter-shared across intersections and executed in a decentralized manner.

For the discrete action space, the critic networks are updated by minimizing the Bellman residual:

$$J_Q(\theta) = \mathbb{E}_{(u_{i,\tau}, a_{i,\tau}, r_\tau, u_{i,\tau+1}) \sim \mathcal{D}} [(Q_\theta(u_{i,\tau}, a_{i,\tau}) - \hat{Q}(u_{i,\tau}, a_{i,\tau}))^2]. \quad (33)$$

where the target value $\hat{Q}$ is calculated as

$$\hat{Q}(u_{i,\tau}, a_{i,\tau}) = r_\tau + \gamma \sum_{a'_{i,\tau+1} \in \mathcal{A}_{\text{loc}}} \pi(a'_{i,\tau+1} | u_{i,\tau+1}) [\bar{Q}_\theta(u_{i,\tau+1}, a'_{i,\tau+1}) - \alpha_e \log \pi(a'_{i,\tau+1} | u_{i,\tau+1})]. \quad (34)$$

The policy network is optimized by minimizing the following objective:

$$J_\pi(\phi) = \mathbb{E}_{u_{i,\tau} \sim \mathcal{D}} \left[\sum_{a_{i,\tau} \in \mathcal{A}_{\text{loc}}} \pi_\phi(a_{i,\tau} | u_{i,\tau}) (\alpha_e \log \pi_\phi(a_{i,\tau} | u_{i,\tau}) - Q_\theta(u_{i,\tau}, a_{i,\tau})) \right]. \quad (35)$$

The reward function is designed to balance queue length reduction and throughput maximization. For each intersection $i \in V$, the local reward at step τ is defined as

$$r_{i,\tau} = \alpha N_{i,\tau} - \beta L_{i,\tau}, \quad (36)$$

where $L_{i,\tau}$ and $N_{i,\tau}$ denote the total queue length and the number of departing vehicles at intersection i during the current interval, respectively. The global environment reward is the sum of all local rewards:

$$r_\tau = \sum_{i \in V} r_{i,\tau}. \quad (37)$$

In our experiments, we set $\alpha = 1$ and $\beta = 0.1$.

From a traffic-flow perspective, the adopted observation and reward design also admit a physical interpretation. The observation space is designed to capture not only local traffic demand and operating conditions at each intersection, but also the interaction environment induced by adjacent roads and neighboring intersections. Therefore, the model input is

intended to provide a compact representation of local congestion states and inter-intersection propagation conditions, rather than serving as a purely generic black-box feature set.

Although the reward function is expressed in a simple form, its control objective is consistent with traffic-flow principles. Specifically, it encourages effective vehicle discharge while suppressing persistent queue accumulation. In this sense, the reward is aligned with the physical objective of reducing congestion growth, mitigating spillback risk, and improving coordinated traffic progression across neighboring intersections. Therefore, the adopted design can be viewed as a learning-based approximation with traffic-oriented physical motivation, rather than a purely heuristic black-box formulation.

In summary, MPNNLight maps multi-intersections into a graph, then employs AWSformer-based message passing to capture global coordination and fuse graph and node features for robust signal control.

The pseudocode of the MPNNLight algorithm is presented in Algorithm 1.

Algorithm 1 Training Procedure of MPNNLight

```

1: Input: traffic environment  $\mathcal{E}$ ; directed road network graph  $G = (V, E)$ ; training horizon  $T$ ; number of
   message-passing layers  $K$ ; replay buffer  $\mathcal{D}$ ; SAC hyper-parameters.
2: Output: trained policy network  $\pi$  and critic networks  $Q_1, Q_2$ .
3: Initialize the policy network  $\pi$ , critic networks  $Q_1, Q_2$ , and target critic networks  $\bar{Q}_1, \bar{Q}_2$ .
4: Initialize replay buffer  $\mathcal{D}$ .
5: for each training episode do
6:   Reset the environment  $\mathcal{E}$  and obtain the initial observation  $o_\tau$ .
7:   for  $\tau = 1, \dots, T$  do
8:     (1) Extract local traffic features.
9:     For each intersection  $i \in V$ , obtain the vehicle-state matrix  $X_{i,\tau}$  and the statistical state vector  $s_{i,\tau} = [N_{i,\tau}^V, S_{i,\tau}^V, P_{i,\tau}^L]$ .
10:    Compute the convolutional feature  $c_{i,\tau} = f_{\text{conv}}(X_{i,\tau})$  and the local representation  $h_{i,\tau} = [c_{i,\tau} \parallel \text{LN}(s_{i,\tau})]$ .
11:    (2) Construct graph attributes.
12:    Build the initial node feature for each intersection as  $f_{i,\tau}^{(0)} = [N_{i,\tau}^V, S_{i,\tau}^V]$ .
13:    Build the edge feature for each directed road  $(i, j) \in E$  as  $e_{ij,\tau} = [L_{ij}, S_{ij,\tau}, N_{ij,\tau}]$ .
14:    Form the attributed graph at decision step  $\tau$  as  $\mathcal{G}_\tau = (V, E, \{f_{i,\tau}^{(0)}\}_{i \in V}, \{e_{ij,\tau}\}_{(i,j) \in E})$ .
15:    (3) Message passing with AWSformer.
16:    Initialize node embeddings as  $f_{i,\tau}^{(0)}$  for all  $i \in V$ .
17:    for  $t = 1, \dots, K$  do
18:      For each intersection  $i \in V$ , compute messages from neighboring intersections  $j \in \mathcal{N}(i)$  using
        AWSformer based on node features and edge features.
19:      Aggregate the incoming messages and update the node embedding to obtain  $f_{i,\tau}^{(t)}$ .
20:    end for
21:    (4) Readout for global traffic context.
22:    Apply a readout operation over  $\{f_{i,\tau}^{(K)}\}_{i \in V}$  to obtain the global graph representation  $g_{\text{global},\tau}$ .
23:    (5) Fuse local and global features.
24:    For each intersection  $i \in V$ , fuse the local representation  $h_{i,\tau}$  and the global representation  $g_{\text{global},\tau}$  to
      obtain the decision feature  $u_{i,\tau}$ .
25:    (6) Generate control actions.
26:    For each intersection  $i \in V$ , compute phase logits  $z_{i,\tau} = \text{LN}_U(u_{i,\tau})$ .
27:    Obtain the policy distribution over the local action space  $\mathcal{A}_{\text{loc}} = \{1, 2, 3, 4\}$  by  $\pi(a_{i,\tau} \mid u_{i,\tau}) = \text{Softmax}(z_{i,\tau})$ .
28:    Sample the action for each intersection as  $a_{i,\tau} \sim \pi(\cdot \mid u_{i,\tau})$ .
29:    Form the joint action  $\mathbf{a}_\tau = \{a_{i,\tau}\}_{i \in V}$ .
30:    (7) Interact with the environment.
31:    Execute  $\mathbf{a}_\tau$  in  $\mathcal{E}$ , receive reward  $r_\tau$ , and observe the next observation  $o_{\tau+1}$ .
32:    (8) Store transition.
33:    Store  $(o_\tau, \mathbf{a}_\tau, r_\tau, o_{\tau+1})$  in replay buffer  $\mathcal{D}$ .
34:    (9) Update networks using SAC.
35:    Sample a minibatch of transitions from  $\mathcal{D}$  and convert each joint transition into per-agent samples
       $(u_{i,\tau}, a_{i,\tau}, r_\tau, u_{i,\tau+1})$ .
36:    Update the critic networks  $Q_1, Q_2$  by minimizing the Bellman backup loss.
37:    Update the policy network  $\pi$  by maximizing the entropy-regularized expected return.
38:    Softly update the target critic networks  $\bar{Q}_1, \bar{Q}_2$ .
39:    Set  $o_\tau \leftarrow o_{\tau+1}$ .
40:   end for
41: end for

```

4. Experiments

We conduct traffic signal control experiments using the SUMO simulator to evaluate the performance of the proposed MPNNLight method. SUMO provides a realistic and flexible environment for modeling both synthetic and real-world urban traffic systems, enabling efficient simulation of multi-intersection interactions under different traffic conditions. By integrating the traffic simulator with the reinforcement learning framework, the proposed model can interact with the environment and progressively optimize signal control policies based on observed traffic states. During the simulation, real-time traffic states are continuously collected, and the agent executes control actions at each decision step. Each simulation lasts for 3600 s with a control interval of 10 s, and the vehicle length is set to 5 m. To comprehensively assess the effectiveness of MPNNLight, we first conduct experiments on synthetic 2×2 intersection datasets under controlled traffic settings, and then further evaluate the proposed method on larger real-world traffic networks including Jinan 3×4 and Hangzhou 4×4 to verify its scalability and generalization ability.

4.1. Experimental Results on Synthetic Datasets

To evaluate the proposed method in a clear and controlled manner, we first conduct experiments on eight synthetic datasets (Dataset-0 to Dataset-7) generated on a fixed 2×2 intersection topology. The purpose of starting from this setting is not merely to simplify the problem, but to isolate the effect of traffic-demand variation from other confounding factors. Since all eight datasets share the same network structure, differences in performance can be attributed mainly to changes in traffic intensity and flow patterns rather than to variations in road topology, intersection layout, or connectivity. This makes the 2×2 setting particularly suitable for examining whether the proposed message-passing mechanism can adapt consistently to different traffic conditions.

Another important reason for using a fixed 2×2 topology in the first-stage evaluation is that it provides a rigorous testbed for analyzing the stability of coordination behavior. In larger real-world networks, performance is jointly affected by many factors, such as irregular road geometry, heterogeneous intersection roles, and uneven connectivity across regions. These factors may obscure whether a method truly learns effective inter-intersection coordination or merely benefits from specific structural properties of a particular network. By contrast, experiments on a unified synthetic topology allow us to focus on the intrinsic capability of the model to capture spatial dependencies, propagate informative messages, and maintain robust control performance as congestion levels increase.

The eight datasets are designed to cover a wide range of traffic regimes, from light traffic to highly congested conditions, with vehicle counts ranging from 600 to 5662. Some datasets reflect relatively regular traffic demand, while others correspond to deliberately constructed high-pressure scenarios. Such a design enables us to systematically examine how the proposed method behaves under progressively more challenging traffic loads within the same structural environment. Each dataset simulates 3600 s of traffic, and each vehicle is represented as (o, t, d) , where o , t , and d denote the origin, departure time, and destination, respectively. The traffic volume of each dataset is summarized in Table 1.

It should be noted that this synthetic 2×2 evaluation is intended as the first step of the experimental study rather than the final validation. After verifying the effectiveness of the proposed method under controlled topology-consistent settings, we further evaluate it on larger real-world traffic networks in the next section, including three Jinan 3×4 networks and two Hangzhou 4×4 networks. This two-stage experimental design allows us to first establish the method's behavior under controlled traffic variations and then assess its scalability and practical effectiveness on more complex urban road networks.

Table 1. Traffic volume of each dataset.

Dataset	Traffic Volume (Vehicles)
Dataset-0	5662
Dataset-1	5600
Dataset-2	4600
Dataset-3	3600
Dataset-4	2600
Dataset-5	1600
Dataset-6	5560
Dataset-7	600

To validate the effectiveness of the AWSformer-based MPNN architecture in MPNN-Light for multi-intersection traffic signal control, we compare it with nine baseline methods, including approaches that do not employ graph neural networks as well as several representative graph-based models proposed in prior work. This comparison aims to demonstrate the superiority of the proposed architecture in coordinating traffic signals across multiple intersections. Among them, FixedTime and IDLight serve as non-graph baselines, while the remaining models replace the AWSformer-based MPNN component in MPNNLight with different types of graph neural networks to extract spatial features of the road network. The naming rule is to concatenate the name of the graph neural network algorithm used by the model with “Light”. For example, “GCNLight” refers to a model that uses GCN as its graph neural network architecture. A brief introduction of each model is as follows:

- FixedTime: A traditional method which uses pre-defined signal plans, independent of real-time traffic conditions.
- IDLight: A method which controls each intersection independently without using graph neural networks for global feature extraction.
- GCNLight: A method which incorporates GCN to model spatial correlations between intersections.
- GATLight: A method which utilizes GAT to adaptively weight neighboring intersections based on attention mechanisms.
- GATLight++: A method which is based on GATv2 [36] and uses refined attention and feature aggregation strategies.
- GINLight: A method which applies Graph Isomorphism Networks (GINs) [37] to capture more expressive node-level representations.
- GraphSAGELight: A method which aggregates neighbor information using the Graph sample and aggregate (GraphSAGE) [38] framework for inductive graph learning.
- GraphEdgeConvLight: A method that employs edge convolution to model pairwise relationships between connected nodes in the graph structure [39].
- GraphTransformerLight: A method which integrates Transformer-based attention for long-range dependency modeling across nodes in the graph.

We conduct experiments on 8 datasets using these algorithms. Figure 3 and Figure 4 present the training curves on Dataset-5 and Dataset-7, respectively. From the two figures, it can be observed that MPNNLight consistently demonstrates more desirable convergence behavior than the competing methods in terms of convergence speed, final duration, and training stability. On Dataset-5, although most methods exhibit a rapid decrease in duration during the initial training stage, several baselines, such as GATLight++, GINLight, and GraphTransformerLight, show pronounced initial spikes and relatively large fluctuations, indicating that their optimization process is less stable and more sensitive to early-stage parameter updates. In contrast, MPNNLight reduces the duration more smoothly after a short warm-up phase and quickly enters a stable convergence regime. Furthermore,

after the early descending stage, its curve remains below those of almost all competing methods for most of the training process, suggesting that the proposed method can achieve higher efficiency with fewer unnecessary oscillations. A similar trend can also be observed on Dataset-7. Although the overall fluctuations on this dataset are weaker than those on Dataset-5, MPNNLight still reaches a low-duration region faster than the other methods and maintains the smallest final convergence value. By comparison, IDLight converges more slowly and remains at a relatively high duration level throughout the training process, while the other lightweight graph learning methods, including GCNLight, GATLight, GINLight, GraphSAGELight, GraphEdgeConvLight, and GraphTransformerLight, gradually stabilize but still fail to attain the final convergence level achieved by MPNNLight. In addition, several baseline methods still exhibit occasional local spikes or oscillatory behavior even in the middle or later stages of training, whereas the curve of MPNNLight remains comparatively smooth, reflecting stronger robustness and a more reliable optimization trajectory. Overall, the results on both Dataset-5 and Dataset-7 consistently verify that MPNNLight achieves a better trade-off among fast convergence, low final duration, and stable training dynamics, thereby demonstrating its superior efficiency and robustness over the competing lightweight graph learning methods.

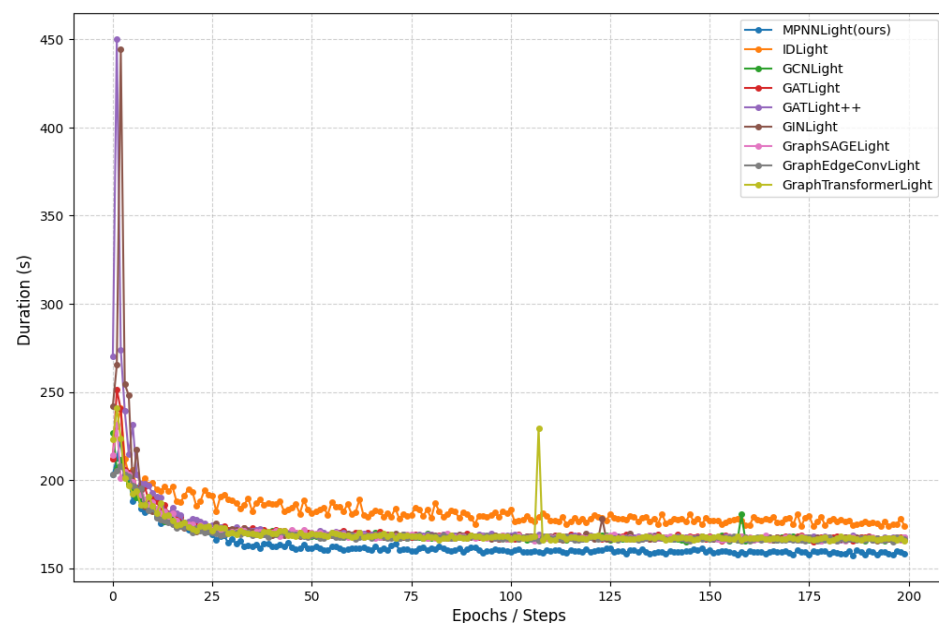


Figure 3. Training curve on Dataset-5 in terms of average travel time.

Tables 2 and 3 summarize the average travel time and average waiting time of vehicles across eight datasets. Overall, MPNNLight achieves the best performance among all compared methods, yielding the lowest average travel time of 222.89 s and the lowest average waiting time of 79.69 s. More importantly, this advantage is consistent across all eight datasets rather than being confined to a few specific cases. MPNNLight ranks first on every dataset for both metrics, indicating that the proposed model can simultaneously improve global traffic efficiency and reduce vehicle-level delays under diverse traffic conditions. Such consistent superiority across datasets with different scales and flow patterns suggests that the proposed framework captures stable and transferable coordination mechanisms instead of overfitting to a particular traffic scenario.

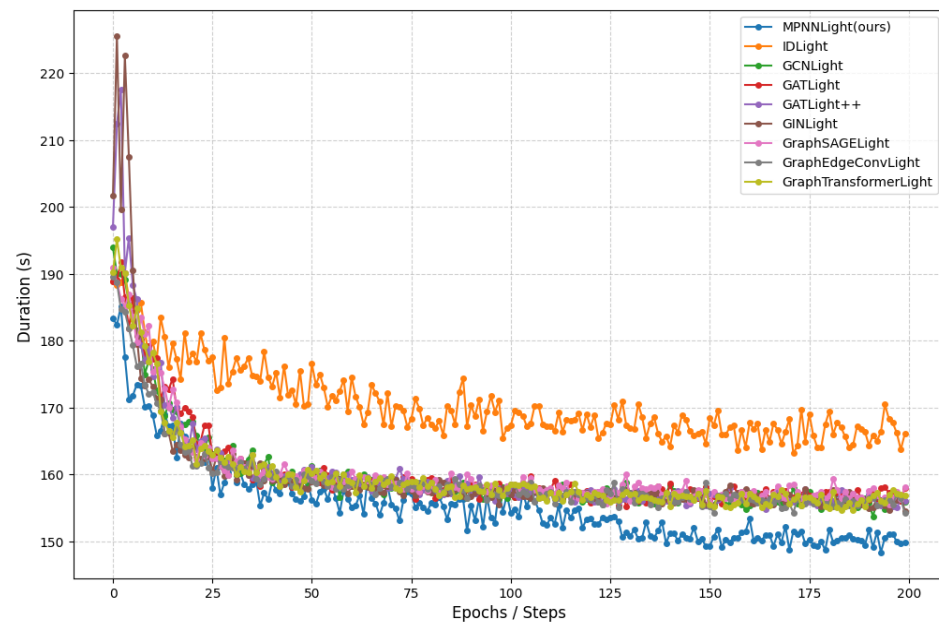


Figure 4. Training curve on Dataset-7 in terms of average travel time.

Table 2. Average travel time (s) of vehicles across eight datasets.

Method	Dataset-0	Dataset-1	Dataset-2	Dataset-3	Dataset-4	Dataset-5	Dataset-6	Dataset-7	Average
FixedTime	298.95	652.81	617.06	482.58	306.48	220.46	114.01	189.84	360.27
IDLight	160.29	480.43	400.67	254.40	197.90	167.09	98.75	164.73	240.53
GCNLight	136.07	462.25	399.46	242.54	191.44	157.54	92.57	159.66	230.19
GATLight	131.12	470.79	379.41	242.38	187.68	158.19	92.50	158.80	227.61
GATLight++	150.48	473.91	381.70	240.23	190.27	157.96	93.14	158.93	230.83
GINLight	131.33	459.42	384.00	236.81	187.39	157.61	91.50	159.15	225.90
GraphSAGELight	132.63	461.00	380.62	240.46	197.49	158.02	91.12	159.24	227.57
GraphEdgeConvLight	132.23	459.40	377.37	237.63	189.47	157.37	92.50	158.50	225.56
GraphTransformerLight	131.05	458.92	380.43	238.87	189.41	157.99	93.75	158.04	226.06
MPNNLight (ours)	130.81	457.84	376.02	236.50	186.76	157.31	88.50	149.37	222.89

Table 3. Average waiting time (s) of vehicles across eight datasets.

Method	Dataset-0	Dataset-1	Dataset-2	Dataset-3	Dataset-4	Dataset-5	Dataset-6	Dataset-7	Average
FixedTime	58.67	339.88	219.54	112.44	56.39	36.02	21.00	33.04	109.62
IDLight	34.08	284.05	196.96	89.45	47.49	27.68	13.57	22.93	89.53
GCNLight	29.34	276.32	192.49	76.04	44.68	25.08	7.62	20.31	83.99
GATLight	28.95	273.10	193.83	75.64	44.99	24.07	8.75	21.27	83.83
GATLight++	50.57	278.02	188.87	76.42	45.12	24.58	9.12	21.49	86.77
GINLight	28.50	273.26	191.91	76.44	44.25	24.86	9.13	20.54	83.61
GraphSAGELight	28.63	280.31	191.98	75.84	44.60	24.30	8.99	21.02	84.46
GraphEdgeConvLight	28.61	271.05	195.61	76.27	44.56	25.58	10.29	20.27	84.03
GraphTransformerLight	28.78	273.17	189.18	76.49	45.02	26.56	7.38	20.29	83.36
MPNNLight (ours)	26.35	267.81	183.46	73.15	42.06	21.84	5.01	17.83	79.69

A clear trend observed from both tables is that IDLight, although significantly outperforming the FixedTime baseline, remains inferior to all graph-based methods in most cases. Since IDLight controls each intersection independently and does not explicitly model the topological dependencies among adjacent intersections, its limited performance indicates that local decision-making alone is insufficient for complex urban traffic networks, where congestion can propagate spatially across multiple intersections. In contrast, all graph-based baselines, including GCNLight, GATLight, GINLight, GraphSAGELight, GraphEdgeConvLight, and GraphTransformerLight, consistently achieve lower travel time and waiting time than IDLight. This comparison confirms the necessity of incorpo-

rating road-network topology into the control process and highlights the importance of inter-intersection coordination for alleviating traffic congestion.

Compared with the existing GNN-based baselines, MPNNLight further delivers consistent improvements on all datasets, indicating that the performance gain is not merely due to the use of graph neural networks, but also to the superior design of the proposed architecture. Although the competing graph-based methods are capable of modeling spatial correlations to some extent, their aggregation mechanisms are still limited in capturing long-range dependencies, heterogeneous interaction strengths, and dynamic traffic propagation patterns across the network. By contrast, the AWSformer-based design in MPNNLight enables more expressive message aggregation and feature interaction, allowing the model to better identify critical traffic dependencies and integrate both local and global contextual information. As a result, MPNNLight learns more effective control policies and achieves lower travel time as well as lower waiting time in all traffic scenarios.

Another important observation is that the superiority of MPNNLight remains evident under different types of control difficulty. On Dataset-1 and Dataset-2, where all methods exhibit relatively large travel and waiting times, MPNNLight still attains the best results, with average travel times of 457.84 s and 376.02 s and average waiting times of 267.81 s and 183.46 s, respectively. This demonstrates that the proposed model remains effective under relatively heavy traffic demand and more complex traffic dynamics. A more representative contrast can be observed between Dataset-6 and Dataset-7. Although Dataset-6 has a substantially larger traffic volume of 5560 vehicles, its traffic flows are more evenly distributed across directions, making the signal control task relatively easier from a global coordination perspective. Accordingly, MPNNLight achieves an average travel time of 88.50 s and an average waiting time of only 5.01 s on this dataset. In contrast, Dataset-7 contains only 600 vehicles, but its traffic demand is concentrated in specific directions, resulting in a highly imbalanced flow pattern. Consequently, even MPNNLight yields a markedly higher average travel time of 149.37 s and average waiting time of 17.83 s on Dataset-7. This comparison indicates that traffic control difficulty is determined not only by the total traffic volume, but also by the spatial distribution of traffic demand. The strong performance of MPNNLight on both datasets therefore demonstrates its robustness in both high-volume but balanced scenarios and low-volume yet highly anisotropic traffic conditions.

It is also noteworthy that the gains of MPNNLight are consistently reflected in both evaluation metrics. Average travel time characterizes the overall efficiency of traffic movement through the network, whereas average waiting time directly reflects the delay experienced by vehicles at intersections. In practical traffic control, improvements in one metric may sometimes come at the expense of the other. However, MPNNLight simultaneously reduces both metrics across all datasets, indicating that it does not merely improve traffic flow in a short-sighted or localized manner, but instead learns a more balanced control strategy that enhances system-level throughput while effectively suppressing queueing delay and idle waiting.

This conclusion is further supported by the representative case studies shown in Figures 5–7. On Dataset-4, MPNNLight achieves the lowest average travel time of 186.76 s and the lowest average waiting time of 42.06 s, corresponding to reductions of 39.06% and 25.41% relative to FixedTime, respectively. Even compared with the strongest competing baseline, MPNNLight still reduces travel time by 0.63 s relative to GINLight (187.39 s) and waiting time by 2.19 s relative to GINLight (44.25 s). On Dataset-6, MPNNLight achieves 88.50 s in travel time and 5.01 s in waiting time, corresponding to reductions of 22.38% and 76.14%, respectively. Compared with the best baseline, it further reduces travel time by 2.62 s relative to GraphSAGELight (91.12 s) and waiting time by 2.37 s relative to GraphTransformerLight (7.38 s). On Dataset-7, MPNNLight obtains the lowest average

travel time of 149.37 s and average waiting time of 17.83 s, with corresponding reductions of 21.32% and 46.04% relative to FixedTime. Compared with the best competing results, it further improves travel time by 8.67 s relative to GraphTransformerLight (158.04 s) and waiting time by 2.44 s relative to GraphEdgeConvLight (20.27 s). These representative examples further demonstrate that the proposed method maintains clear advantages under moderate, balanced, and highly imbalanced traffic scenarios.

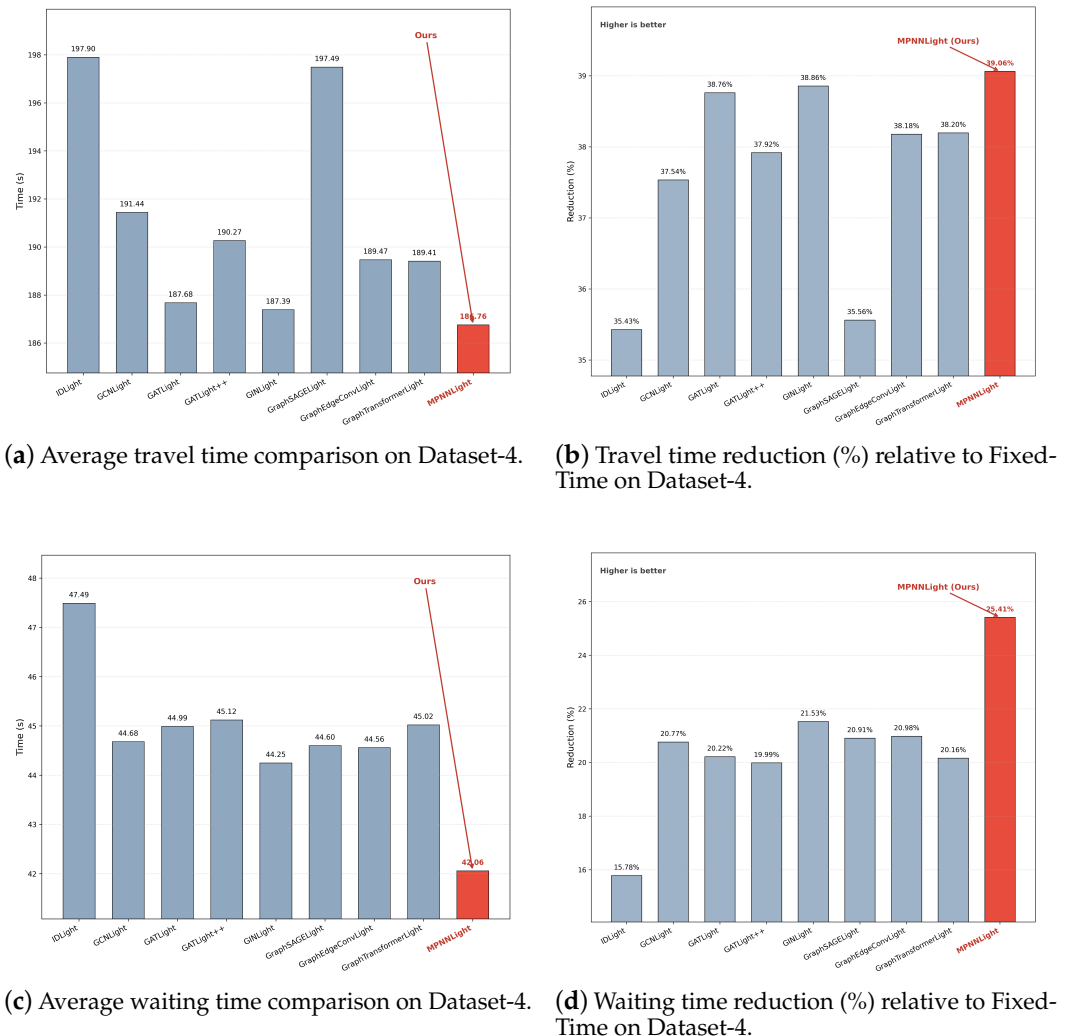


Figure 5. Performance comparison on Dataset-4. (a,b) show the average travel time and its percentage reduction relative to the FixedTime baseline. (c,d) illustrate the waiting time and its corresponding percentage reduction, where higher reduction values indicate better performance.

4.2. Experimental Results on Real-World Datasets

To further validate the effectiveness and practical applicability of the proposed MPNN-Light algorithm in real-world multi-intersection traffic signal control scenarios, we conduct experiments on five real-world traffic datasets collected from Jinan and Hangzhou. Evaluating models on real-world datasets is of critical importance, as it enables a more faithful assessment of model performance under complex, dynamic, and inherently stochastic traffic conditions, which are often inadequately represented in synthetic environments.

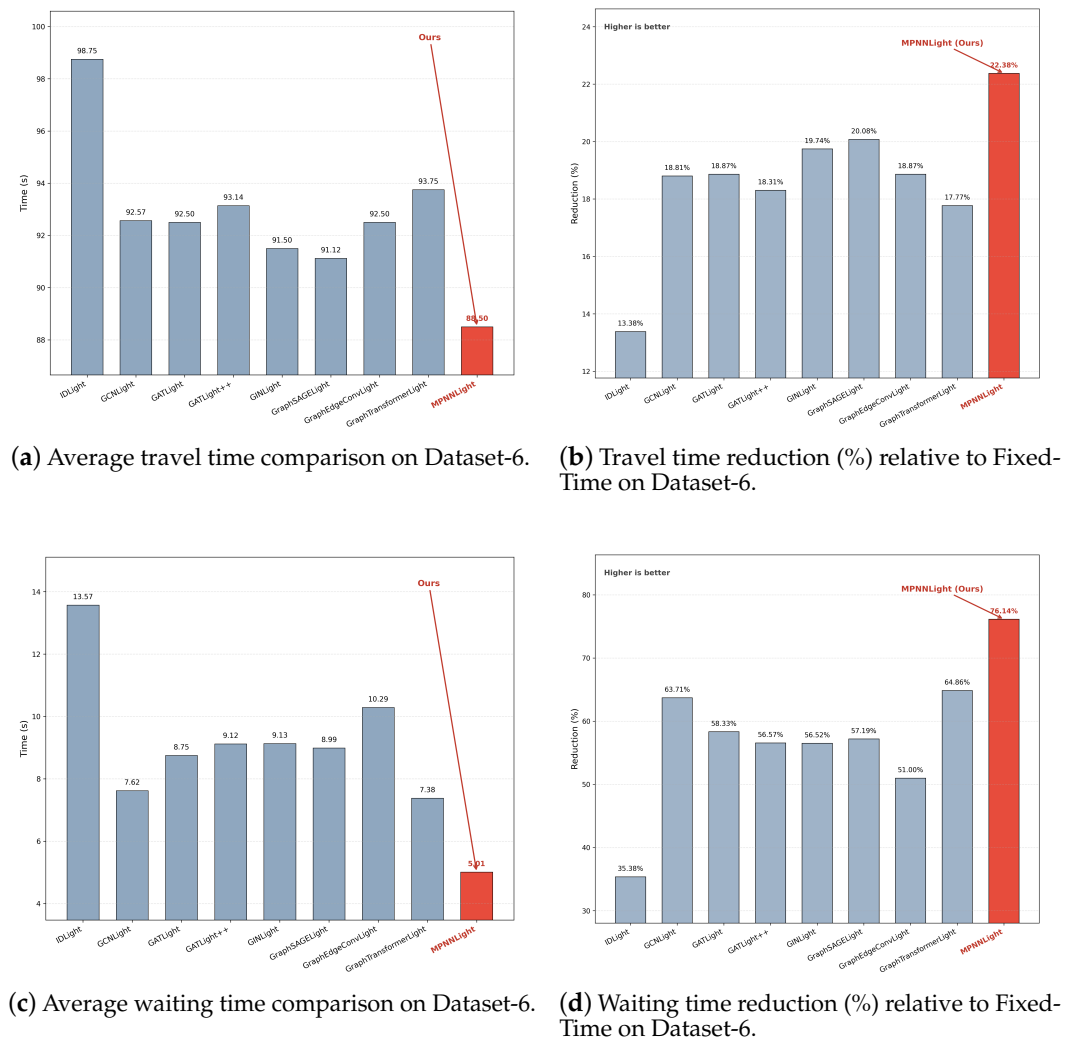


Figure 6. Performance comparison on Dataset-6. (a,b) show the average travel time and its percentage reduction relative to the FixedTime baseline. (c,d) illustrate the waiting time and its corresponding percentage reduction, where higher reduction values indicate better performance.

Specifically, the Jinan dataset corresponds to a 3×4 urban road network with 12 signalized intersections, while the Hangzhou dataset is constructed on a 4×4 network consisting of 16 intersections. These networks differ not only in scale but also in traffic demand distributions and spatial configurations, thereby providing a diverse and challenging testbed for evaluating the robustness and scalability of the proposed method. The selected datasets further exhibit varying levels of traffic congestion, ranging from moderate demand to highly saturated conditions, which facilitates a comprehensive evaluation across different operating regimes.

Figure 8 presents both the schematic layouts and satellite views of the studied regions. Figure 8a illustrates the Jinan network, including its abstracted intersection topology and real satellite imagery, while Figure 8b shows the corresponding representations for the Hangzhou network. These visualizations clearly demonstrate the structural complexity, irregular road geometries, and spatial heterogeneity inherent in real-world traffic systems, which pose significant challenges for learning-based control methods.

The traffic volumes of the selected datasets are summarized in Table 4. As shown in Table 4, the datasets cover a wide range of traffic scales, reflecting diverse traffic intensities encountered in real urban environments. Such diversity is essential for systematically

evaluating the generalization capability of the proposed MPNNLight algorithm across different traffic conditions, and for demonstrating its potential for real-world deployment.

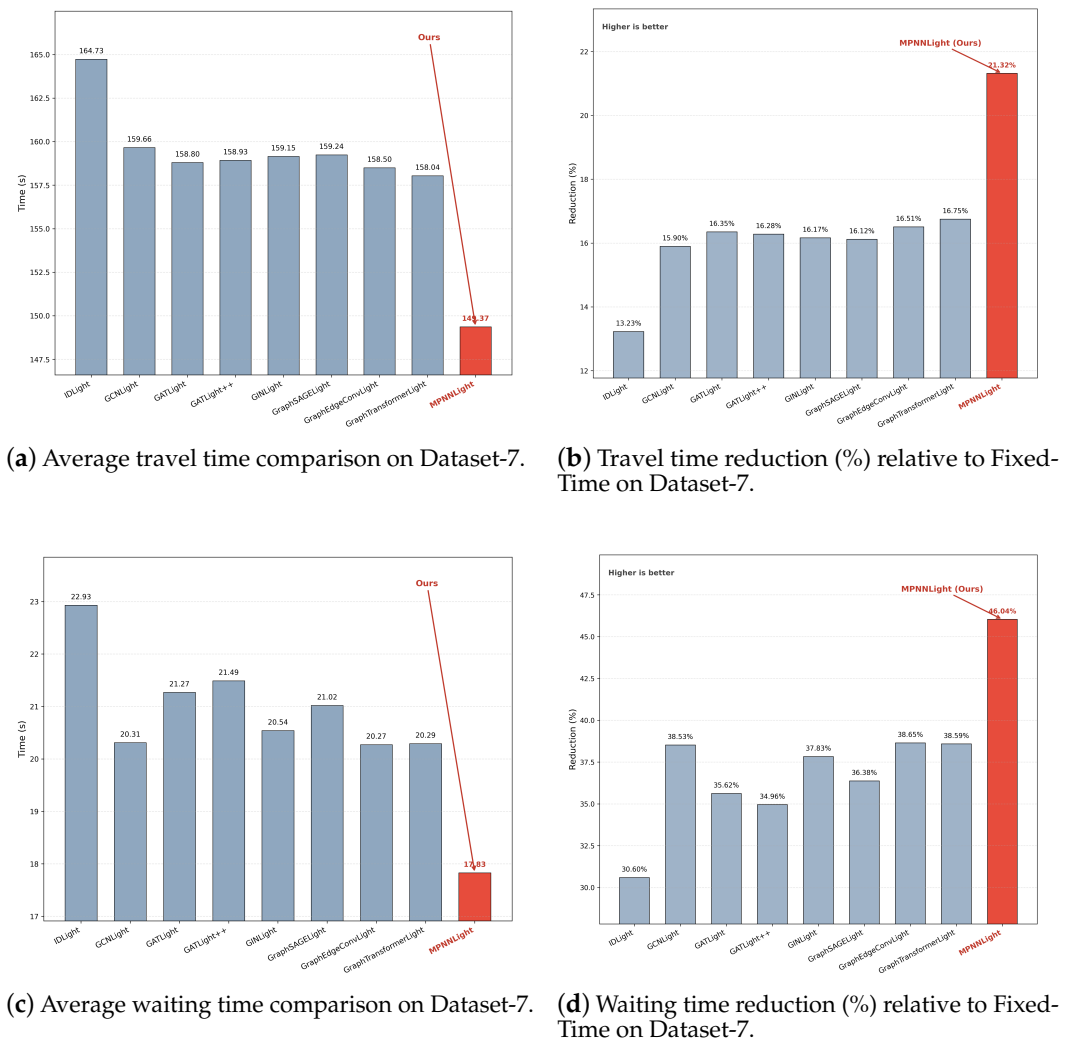


Figure 7. Performance comparison on Dataset-7. (a,b) show the average travel time and its percentage reduction relative to the FixedTime baseline. (c,d) illustrate the waiting time and its corresponding percentage reduction, where higher reduction values indicate better performance.

Table 4. Traffic volume of selected real-world datasets.

Dataset	Traffic Volume (Vehicles)
Jinan-1	6295
Jinan-2	4365
Jinan-3	5494
Hangzhou-1	2983
Hangzhou-2	6984

We further compare MPNNLight with a series of representative reinforcement learning-based traffic signal control methods, including FixedTime, MPLight, PressLight, CoLight, IDLight, GCNLight, GATLight, GATLight++, GINLight, GraphSAGELight, GraphEdgeConvLight, and GraphTransformerLight. Compared with the previous experiments on synthetic datasets, we additionally include MPLight [40], PressLight [41], and CoLight [13] in the real-world evaluation. The reason for introducing these three methods is that they are widely recognized as strong and representative baselines in real-world traffic signal

control, especially under large-scale or partially coordinated urban traffic scenarios. In particular, PressLight and MPLight are pressure-based methods that have demonstrated strong effectiveness in practical deployments, while CoLight explicitly models inter-intersection coordination through attention-based communication. Therefore, comparing against these methods on real-world datasets provides a more rigorous validation of whether the proposed MPNNLight can maintain superior performance under realistic road topologies, heterogeneous traffic flows, and complex spatial dependencies.

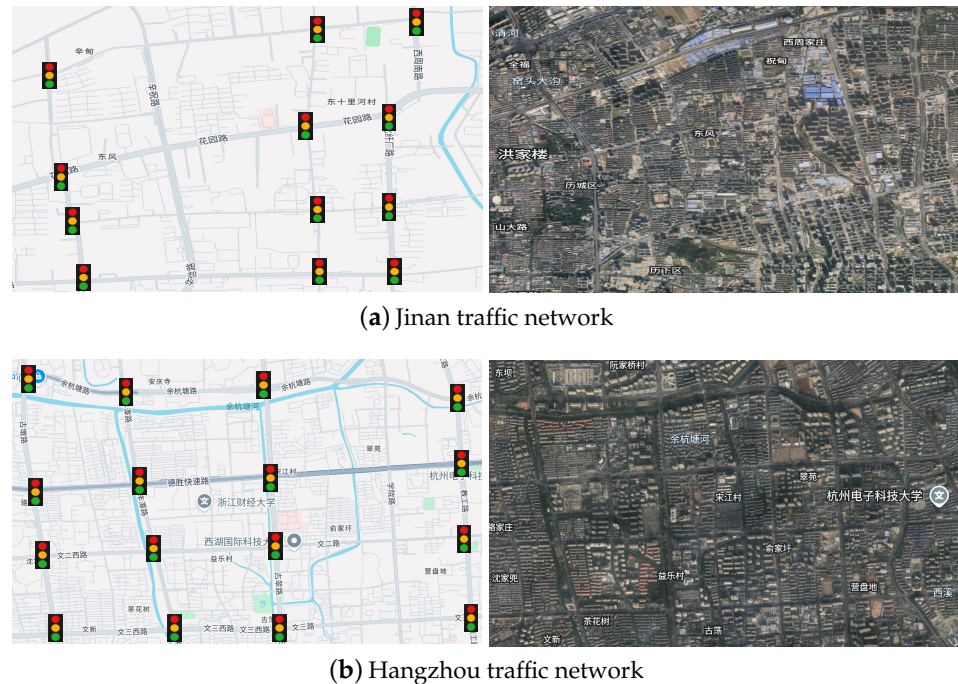


Figure 8. Schematic layouts and satellite views of the real-world traffic networks. (a) Jinan: intersection topology and satellite image. (b) Hangzhou: intersection topology and satellite image.

Figure 9 illustrates the training curves of average travel time for different methods on the five real-world multi-intersection datasets. As shown in Figure 9, MPNNLight consistently achieves faster convergence, lower average travel time, and more stable training behavior across all five real-world datasets. On the three Jinan datasets, several baseline methods exhibit obvious fluctuations in the early training stage and remain at relatively high travel times after convergence. In contrast, MPNNLight quickly reduces the average travel time and stabilizes at a lower level, indicating its stronger ability to model inter-intersection dependencies under complex traffic conditions. A similar tendency can be observed on the two Hangzhou datasets. Compared with the representative baselines, MPNNLight maintains a smoother optimization trajectory and converges to lower travel times, while several competing methods still show noticeable oscillations or inferior final performance. These results demonstrate that the proposed model has better robustness and optimization capability in large-scale real-world traffic scenarios. For clarity, only representative baselines are plotted in Figure 9, while the complete quantitative comparison is reported in Table 5. Overall, the training curves further confirm the superiority of MPNNLight on both the Jinan and Hangzhou real-world multi-intersection datasets.

Table 5 reports the average travel time achieved by different methods on the five real-world datasets. Overall, MPNNLight obtains the best performance on all datasets, demonstrating its strong robustness and effectiveness in practical multi-intersection traffic signal control scenarios. More specifically, on the three Jinan datasets, MPNNLight achieves average travel times of 271.78 s, 263.81 s, and 255.74 s, respectively, which are consistently

lower than those of all competing methods. On the two Hangzhou datasets, MPNNLight further attains the best results of 276.67 s and 367.68 s, again outperforming all baselines. These results indicate that the proposed method is not only effective on networks with different sizes and topological structures, but also robust under varying traffic demand patterns and congestion levels.

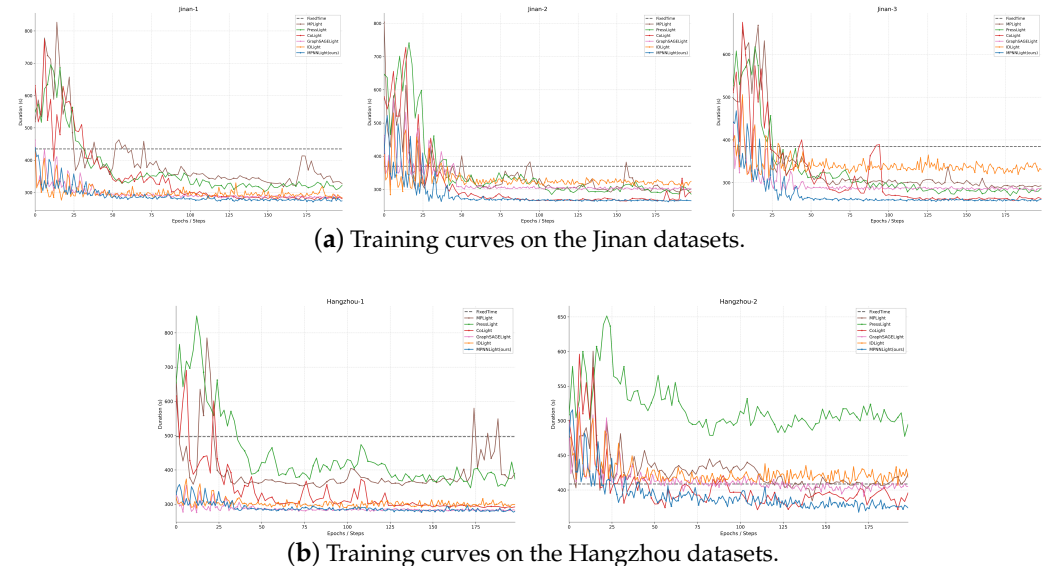


Figure 9. Training curves of average travel time on five real-world multi-intersection datasets. Subfigure (a) shows the results on Jinan-1, Jinan-2, and Jinan-3, while subfigure (b) shows the results on Hangzhou-1 and Hangzhou-2.

A closer examination of the results shows that MPNNLight yields particularly clear advantages over classical and representative baselines. Compared with FixedTime, the proposed method substantially reduces average travel time on all datasets, confirming the necessity of adaptive learning-based control in real-world traffic environments. When compared with MPlight and PressLight, MPNNLight still maintains a clear performance margin across all five datasets. This observation is particularly important because both MPlight and PressLight are strong pressure-based methods that have been shown to perform competitively in urban traffic control tasks. The consistent superiority of MPNNLight suggests that incorporating message passing over the traffic network enables the model to capture richer structural dependencies than pressure-only formulations, thereby leading to more effective coordinated control decisions.

In addition, MPNNLight also outperforms CoLight on every real-world dataset. Since CoLight explicitly models inter-intersection communication, this comparison demonstrates that the proposed message passing framework provides a more effective mechanism for aggregating and propagating traffic information across neighboring intersections. This advantage is especially evident on the Hangzhou datasets, where the road network is larger and spatial interactions are more complex. Under such conditions, effective representation and transmission of network-level traffic states become increasingly critical, and the superior results of MPNNLight indicate its stronger capability in modeling long-range and heterogeneous traffic dependencies.

Compared with other graph-based methods, including GCNLight, GATLight, GAT-Light++, GINLight, GraphSAGELight, GraphEdgeConvLight, and GraphTransformerLight, MPNNLight also achieves the best overall performance. This result verifies that the superiority of MPNNLight does not merely come from using graph neural networks, but rather from its tailored message passing design for traffic signal control. By more effectively integrating

local intersection observations and inter-node interaction information, MPNNLight is able to produce more discriminative traffic state representations, which ultimately translate into better signal control policies. The gains are particularly consistent across datasets with different traffic scales, suggesting that the proposed model possesses strong generalization capability rather than being specialized to a particular network or traffic regime.

Another noteworthy finding is that the improvement of MPNNLight remains stable across both the Jinan and Hangzhou networks, despite their differences in topology, number of intersections, and traffic distributions. This cross-city consistency is highly desirable for real-world deployment, since urban traffic systems often vary significantly in road geometry and demand characteristics. The results therefore provide strong evidence that MPNNLight has good scalability and transferability across heterogeneous real-world traffic environments.

In summary, the results on the five real-world datasets demonstrate that MPNNLight consistently achieves the lowest average travel time among all compared methods. The superiority over pressure-based methods, attention-based coordination methods, and other graph-based baselines confirms the effectiveness of the proposed message passing mechanism in capturing complex spatial interactions in real traffic networks. These findings strongly support the practical applicability of MPNNLight for large-scale real-world multi-intersection traffic signal control.

Table 5. Average travel time (s) of vehicles across five real-world datasets.

Method	Jinan-1	Jinan-2	Jinan-3	Hangzhou-1	Hangzhou-2	Average
FixedTime	434.80	369.65	384.56	497.00	408.66	418.93
MPLight	318.38	291.95	285.69	348.57	384.23	325.76
PressLight	303.01	284.61	269.38	347.42	477.29	336.34
CoLight	278.02	264.97	257.67	289.24	371.32	292.24
IDLight	282.26	308.26	320.12	287.83	406.28	320.95
GCNLight	280.11	290.04	294.36	280.31	376.76	304.32
GATLight	275.32	292.10	295.32	281.30	377.21	304.25
GATLight++	277.39	289.81	263.19	279.09	383.35	298.57
GINLight	276.20	296.45	299.92	277.47	392.13	308.43
GraphSAGELight	278.50	297.23	278.68	278.45	390.01	304.57
GraphEdgeConvLight	272.19	270.36	274.92	277.47	373.06	293.60
GraphTransformerLight	277.72	276.82	261.98	283.42	372.06	294.40
MPNNLight (ours)	271.78	263.81	255.74	276.67	367.68	287.14

To further investigate the effectiveness of the key components in MPNNLight, we conduct ablation experiments on five real-world traffic signal control scenarios, including three Jinan datasets and two Hangzhou datasets. The aim of this study is to evaluate the contribution of each module to the overall performance of the proposed model and to better understand the source of its performance gains.

Specifically, four ablated variants are considered:

- MPNNLight (w/o 2D-DWT) removes the discrete wavelet transform module and the corresponding frequency-aware interaction modeling component to evaluate the contribution of frequency decomposition to traffic dynamics modeling.
- MPNNLight (w/o Edge Features) discards edge-feature information during message passing, with the aim of examining the importance of relational information between connected intersections.
- MPNNLight (w/o Vehicle State) removes the fine-grained vehicle-state matrix branch while retaining only coarse traffic statistics, in order to assess the role of detailed state representation in traffic signal control.

- MPNNLight (w/o Self-Attention) replaces the adaptive attention-based aggregation mechanism with uniform aggregation to investigate whether learnable neighbor weighting is necessary for graph-based representation learning.

Several observations can be drawn from Table 6. The complete MPNNLight achieves the lowest average travel time across all five real-world traffic scenarios, indicating that each component in the proposed architecture contributes positively to the final performance. The ablation results further provide clear evidence for the necessity of the 2D-DWT module, edge features, self-attention mechanism, and vehicle-state branch in modeling complex traffic interactions.

Table 6. Ablation results of MPNNLight on five real-world traffic scenarios in terms of average travel time.

Method	Jinan-1	Jinan-2	Jinan-3	Hangzhou-1	Hangzhou-2	Average
MPNNLight (w/o 2D-DWT)	276.49	311.09	307.77	290.28	414.58	320.04
MPNNLight (w/o Edge Features)	272.06	299.54	305.25	277.17	413.54	313.51
MPNNLight (w/o Vehicle State)	289.50	291.59	298.88	299.00	393.72	314.54
MPNNLight (w/o Self-Attention)	284.52	314.23	311.62	296.81	433.55	328.15
MPNNLight (ours)	271.78	263.81	255.74	276.67	367.68	287.14

The role of the 2D-DWT module is clearly reflected by the consistent increase in average travel time after its removal. Compared with the complete model, eliminating 2D-DWT leads to noticeable degradation in all scenarios, with especially large gaps on Jinan-2, Jinan-3, and Hangzhou-2. This result suggests that frequency-domain decomposition provides important complementary information beyond conventional spatial message passing. By decomposing the spatial message into low-frequency and high-frequency components, the model is able to simultaneously capture globally smooth traffic evolution, local directional perturbations, and abrupt transient fluctuations. Such a design enhances the representation of multi-scale traffic dynamics and improves robustness in complex real-world environments.

The contribution of edge features is also significant. When edge features are removed, the average travel time increases in every dataset, and the degradation becomes more evident in relatively challenging scenarios such as Jinan-2, Jinan-3, and Hangzhou-2. This phenomenon indicates that interactions among intersections cannot be fully characterized by node states alone. In real traffic networks, the propagation of congestion and vehicle flows is strongly influenced by the physical and operational properties of connecting roads. Incorporating edge features therefore enables the model to encode topology-aware relational information and to better distinguish the strength and relevance of messages transmitted through different links.

The importance of the self-attention mechanism is particularly prominent in the ablation study. Removing self-attention causes the most substantial performance drop among all variants, increasing the average travel time from 263.81 to 314.23 on Jinan-2, from 255.74 to 311.62 on Jinan-3, and from 367.68 to 433.55 on Hangzhou-2. These results demonstrate that adaptive attention-based aggregation is crucial for capturing heterogeneous dependencies among neighboring intersections. Without self-attention, the model loses the ability to dynamically assign different importance to incoming messages, which weakens its capacity to model diverse interaction patterns and coordinated traffic dynamics across the network.

The vehicle-state branch also plays an indispensable role in the proposed framework. After removing this branch, the average travel time increases consistently across all scenarios, and the degradation is especially apparent on Jinan-3. This result shows that fine-

grained vehicle-state representations provide useful supplementary information beyond coarse intersection-level statistics. Such detailed traffic descriptors help the model better characterize instantaneous traffic conditions and improve the quality of graph-based interaction modeling. The consistent decline after removing this branch further confirms that integrating microscopic vehicle information is beneficial for enhancing control performance in real-world traffic environments.

Overall, MPNNLight demonstrates superior robustness, scalability, and adaptability, making it a more effective solution for traffic signal control in complex and diverse road-network environments.

5. Conclusions

This paper proposed MPNNLight, a novel graph-based reinforcement learning framework for multi-intersection traffic signal control. Built upon the Message Passing Neural Network paradigm, the proposed method integrates local intersection observations with global network-level information to achieve coordinated decision-making across urban traffic networks. To enhance the expressiveness of inter-intersection communication, we further introduced AWSformer as the message computation module, which combines self-attention and 2D-DWT to jointly model spatial dependencies and frequency-domain traffic variations. In this way, the proposed framework is able to capture not only the topological relationships among adjacent intersections, but also the multi-scale and directional dynamics of traffic propagation.

Extensive experiments on synthetic traffic datasets demonstrated that MPNNLight consistently outperforms a series of representative baseline methods, including both non-graph approaches and several graph neural network-based traffic signal control models. The proposed method achieved the lowest average travel time and average waiting time across all eight benchmark datasets, while also exhibiting faster convergence and more stable optimization behavior during training. These results indicate that the proposed architecture can effectively learn robust coordination strategies under diverse traffic densities and flow distributions, rather than being specialized to a limited set of traffic scenarios.

The effectiveness of MPNNLight was further validated on five real-world datasets collected from Jinan and Hangzhou. Experimental results showed that the proposed method consistently achieved the best performance on all real-world scenarios, demonstrating strong robustness under different road topologies, network scales, and traffic demand patterns. In particular, the superiority of MPNNLight over pressure-based methods, attention-based coordination models, and other graph-based baselines confirms its practical applicability in realistic multi-intersection traffic signal control environments.

In addition, the ablation study further verified the necessity of the major components in the proposed framework. Removing the 2D-DWT module, edge features, self-attention mechanism, or fine-grained vehicle-state branch all led to clear performance degradation, confirming that each module contributes positively to the final control performance. These findings demonstrate that the strong performance of MPNNLight does not arise from a single design choice, but from the effective collaboration of multi-scale frequency decomposition, topology-aware relational modeling, adaptive message aggregation, and detailed traffic state representation.

Overall, the proposed MPNNLight framework provides an effective and scalable solution for coordinated multi-intersection traffic signal control. By combining flexible message passing with spatial–frequency feature modeling, it offers a promising direction for improving both efficiency and robustness in complex urban traffic systems. In future work, we plan to further extend the proposed framework to larger and more heterogeneous road networks, investigate cross-city transfer and online adaptation capabilities, and incorporate

richer real-world constraints such as pedestrian interactions, public transport priority, and mixed autonomous–human driving environments.

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